

Step 1: Plaintext Baseline

Students must record:

- Accuracy
- Precision / Recall / F1
- Plain inference time (1000 nodes)

This is your reference.

Step 2: Per-Batch Measurements

For each batch (60 nodes):

Client:

- Encryption time
- Ciphertext size

Server (Raspberry Pi):

For now test on same devices

- GAT inference time
- Power consumption
- Peak RAM usage

Client:

- Decryption time
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Step 3: Total Graph Results (After 17 Batches)

Compute:

- Total encryption time
- Total inference time
- Total decryption time
- Total end-to-end latency

$T_{total} = T_{enc} + T_{server} + T_{dec}$

Compute energy:

Energy=Power×Time

Report:

- Energy per batch
 - Total energy for 1000 nodes
 - Energy per node
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Step 4: Accuracy After Decryption

Merge all decrypted batches.

Compare:

- Plain accuracy
 - Encrypted accuracy
 - Accuracy drop (if any)
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Step 5: Small Scalability Test (Very Important)

Test batch sizes:

- 30
- 60
- 120

Measure:

- Latency
- Energy
- Memory

This shows batching trade-off.

Final Results to Present in Paper

End-to-end latency

Energy per 1000-node graph

Accuracy comparison

Memory footprint

Batch-size scaling

